3 (a) Total distance = area under the graph
= 39 m (allow ± 2.0 m) A1

Comments:
This part is generally well done by most of the students.

(b) (i) Change in kinetic energy, $\Delta E_k = \frac{1}{2}m\left(v_f^2 - v_i^2\right) = \frac{1}{2}\times 92\times\left(3^2 - 6^2\right) = -1240$ J A1

Comments:
Quite a lot of students have a misconception that change in ke = ½ m $(\Delta v)^2$, which is very alarming. Besides, also quite a number of students also gives the change in ke as initial ke – final ke which by right should be the reverse.

(ii) Change in gravitational pe, $\Delta E_p = mgh = 92\times 9.81\times 1.3 = 1170$ J A1

Comments:
This part is generally well done by most of the students.

(iii) Useful work done = $P\times t = 75\times 8.0 = 600$ J A1

Comments:
This part is generally well done by most of the students.

(c) (i) Useful work done
= change in ke + change in Gpe + work done overcoming frictional force
$\Rightarrow$ 600 = -1240 + 1170 + work done overcoming frictional force M1
$\Rightarrow$ work done overcoming frictional force = 600 +1240 – 1170 = 670 J A1

Comments:
This part is generally badly done by most of the students. Only the more able students were able to obtain the correct answer based on the correct equation.

(ii) average force = $\frac{670}{39} = 17$ N A1

Comments:
Some of the student use the height of the slope instead of the total distance to calculate the average force.

(d) Total resistive force include both ground friction and air resistance. B1
Air resistance will decrease as the speed decreases, and hence total resistive force would not be constant. B1

Comments:
A lot of the students were not able to state that the total resistive force is a combination of the ground friction and the air resistance. Quite a lot of them still do not have the idea of the dependence of air resistance on the speed of the object which is quite alarming.

4 **(a)** For a body moving in <u>uniform circular motion</u>, there is a <u>change in velocity</u> B1
as the direction of motion is changing